2 **Further calculus**	2.1	Derivation and use of Taylor series.	The derivation, for example, of the expansion of $\sin x$ in ascending powers of $(x - \pi)$ up to and including the term in $(x - \pi)^3$.
	2.2	Use of series expansions to find limits.	E.g. $\lim_{x\to 0}\frac{x-\arctan x}{x^3}$, $\lim_{x\to 0}\frac{e^{2x^2}-1}{x^2}$
	2.3	Leibnitz's theorem.	Leibnitz's theorem for differentiating products.
	2.4	L'Hospital's Rule.	The use of derivatives to evaluate limits of indeterminate forms. Repeated applications and/or substitutions may be required. E.g. $\lim_{x\to 0}\frac{2\sin x-\sin 2x}{x-\sin x}$, $\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^x$

Topics	What students need to learn:		
	Content		Guidance
2 **Further calculus** *continued*	2.5	The Weierstrass substitution for integration.	The use of tangent half angle substitutions to find definite and indefinite integrals. E.g. $\int \operatorname{cosec} x \, \mathrm{d}x$, $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{1+\sin x-\cos x} \, \mathrm{d}x$ using $t = \tan \frac{x}{2}$.